

AyushCase — Non-Coder's Guide to Our Technology

A Clear, Simple & Analogy-Based Guide to Everything in Our Healthcare System

1. The Big Picture: How a Web App Works (Restaurant Analogy)

Think of AyushCase like a high-tech modern hospital clinic or restaurant:

1. Frontend (The Dining Hall & Menu): Everything the patient and doctor see, touch, click, and speak to on their screens.
2. Backend (The Kitchen & Chefs): The invisible brain that takes requests, cooks the data, checks rules, and talks to the AI.
3. Database (The Master Filing Cabinet): The ultra-secure digital vault where patient medical records, ABHA IDs, and prescriptions are stored forever.
4. AI Engine (The Expert Medical Assistant): The intelligent assistant that asks health questions, extracts lab numbers, and prepares summaries for the doctor.

2. Every Technology Used — Explained Simply

Ø=ÞàÞ Next.js 14 & React (The Frontend & Backend Engine)

Simple Analogy: "A super-fast, modular Lego building set for websites that pre-cooks pages before you even open them."

- What it actually does: It renders all web pages (Patient Portal, Case Sheets, Dashboard) and handles server requests in one unified framework.
- Why we used it: It makes the website load in milliseconds on mobile phones and computers without any lag.

Ø=ÞàÞ Prisma ORM & Database (The Digital Filing Cabinet)

Simple Analogy: "An ultra-organized librarian that files every patient record with color-coded labels so nothing is ever lost."

- What it actually does: It organizes and links Doctors, Patients, Case History, and Uploaded Documents through strict relationships.
- Why we used it: It protects data integrity and allows us to easily switch from SQLite to massive hospital databases (like PostgreSQL) with zero code rewrite.

Ø=ÞàÞ NextAuth.js & Bcrypt (The Security Guard & Vault)

Simple Analogy: "A biometric security guard at the hospital door who checks doctor badges and encrypts all passwords into unbreakable codes."

- What it actually does: It manages login sessions securely with encrypted JWT tokens and makes sure Doctor A cannot peek into Doctor B's patients.
- Why we used it: It keeps sensitive patient medical data private, secure, and legally protected.

Ø=ÞàÞ Web Speech API (The Built-In Stenographer)

Simple Analogy: "A voice typist built right inside your phone/laptop that listens to Hindi and English without charging any fee."

- What it actually does: It converts patient speech into text in real time right in the browser (supporting Hindi hi-IN and English en-IN).
- Why we used it: It is 100% free, has zero audio delay, and keeps audio strictly on the patient's device without sending private voices to foreign cloud servers.

Ø=ÞàÞ OCR & Entity Extraction (The Intelligent Medical Scanner)

Simple Analogy: "A smart assistant who reads messy doctor prescriptions and lab reports and highlights abnormal test results in bright red."

- What it actually does: It scans uploaded images or PDFs, extracts medicine names, dosages, and compares lab numbers against standard healthy limits.
- Why we used it: Patients don't have to manually type long medicine names, and doctors instantly spot HIGH/LOW abnormal test values.

Ø=ÞàÞ Hybrid Multi-Model AI (Gemini + OpenAI + Offline Knowledge Fallback)

Simple Analogy: "A team of expert AI consultants backed by an encyclopedia book that works even if the hospital internet cables are cut."

- What it actually does: It talks to Google Gemini or ChatGPT when internet is live, but automatically switches to our built-in Ayurvedic knowledge engine if offline.
- Why we used it: The website never crashes, never shows empty screens, and works 100% reliably during live hackathon presentations.

Ø=ÞàÞ Red Flag Emergency Classifier (The Ambulance Alarm)

Simple Analogy: "An automated hospital alarm that rings immediately if a patient reports life-threatening symptoms."

- What it actually does: It scans patient messages for emergency keywords (chest pain, stroke signs, severe breathlessness) and flashes an emergency banner.
- Why we used it: It ensures AI never delays emergency care for critical patients, meeting the highest medical safety standards.

Ø=Þàþ jsPDF (The 1-Click Prescription Printing Press)

Simple Analogy: "An automated digital printer that turns clinical case sheets into official, stamp-ready PDF documents instantly."

- What it actually does: It formats patient history, Prakriti scores, Nadi findings, and medicines into a clean, printable Ayurvedic prescription.
- Why we used it: Patients and doctors can download, print, or share medical passports with other clinics immediately.

Ø=Þàþ Vercel & GitHub (The Worldwide Cloud Network)

Simple Analogy: "A 24/7 global delivery network that keeps the website running across the world and updates it automatically whenever code changes."

- What it actually does: It hosts the entire website on serverless cloud infrastructure with automated database self-healing on cold starts.
- Why we used it: Anyone anywhere can access AyushCase from their mobile or laptop with zero installation required.

3. The Entire 6-Step Clinical Flow in Simple Words

Step 1: Patient Signs Up & Gives Consent

Patient enters basic info (Name, Age, ABHA ID), chooses Hindi or English, and clicks 1-Click Auto-Fill Demo.

Step 2: AI Conversational Health Interview

Patient speaks or types their problem. The AI asks helpful follow-ups (duration, pain triggers, digestion) while watching for emergency red flags.

Step 3: Upload Old Medical Records (OCR)

Patient snaps photos of past prescriptions or blood tests. The system automatically extracts all medicines and highlights abnormal lab numbers.

Step 4: Visual Timeline & AI Clinical Summary

The system arranges past visits and lab reports into a neat timeline and creates a doctor-ready summary with a verification warning.

Step 5: 1-Click Handover to Doctor

Patient clicks "Send to Doctor". The doctor opens the patient profile and sees the full history, saving ~15 minutes of repetitive questioning.

Step 6: Doctor Consultation & Official Prescription

Doctor performs physical examination (Pulse/Tongue), reviews the pre-filled case sheet, confirms diagnosis, and prints the official prescription!